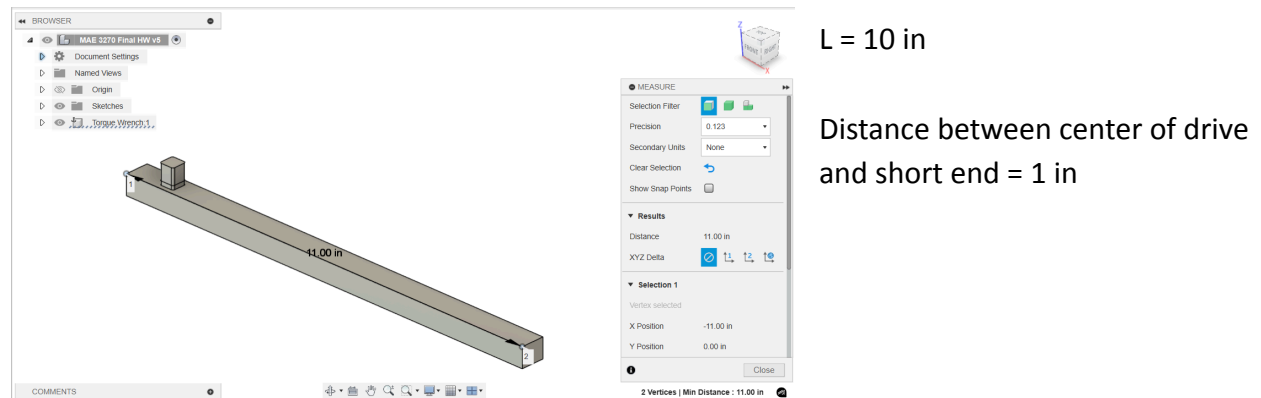
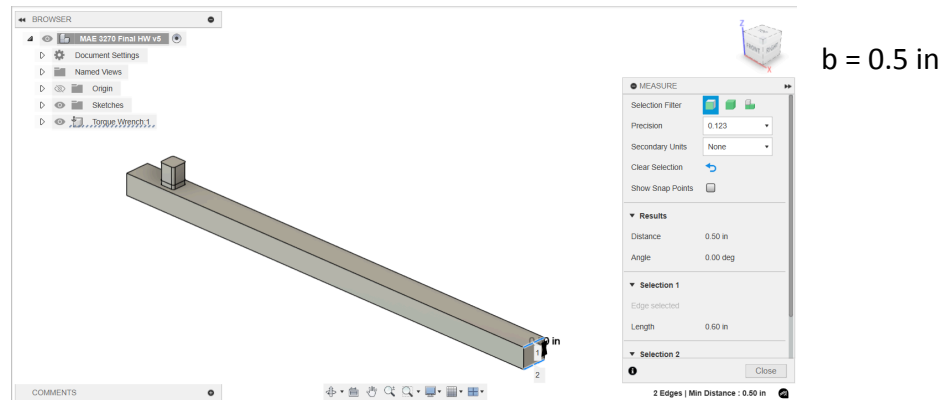
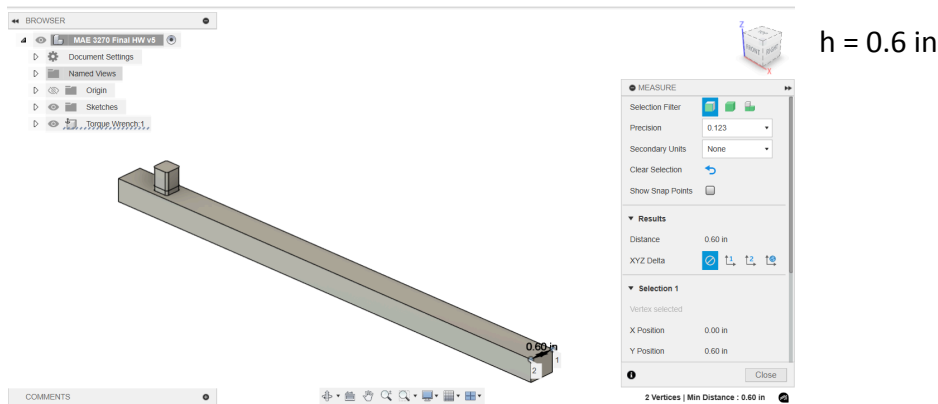


MAE 3270 Final HW

Results

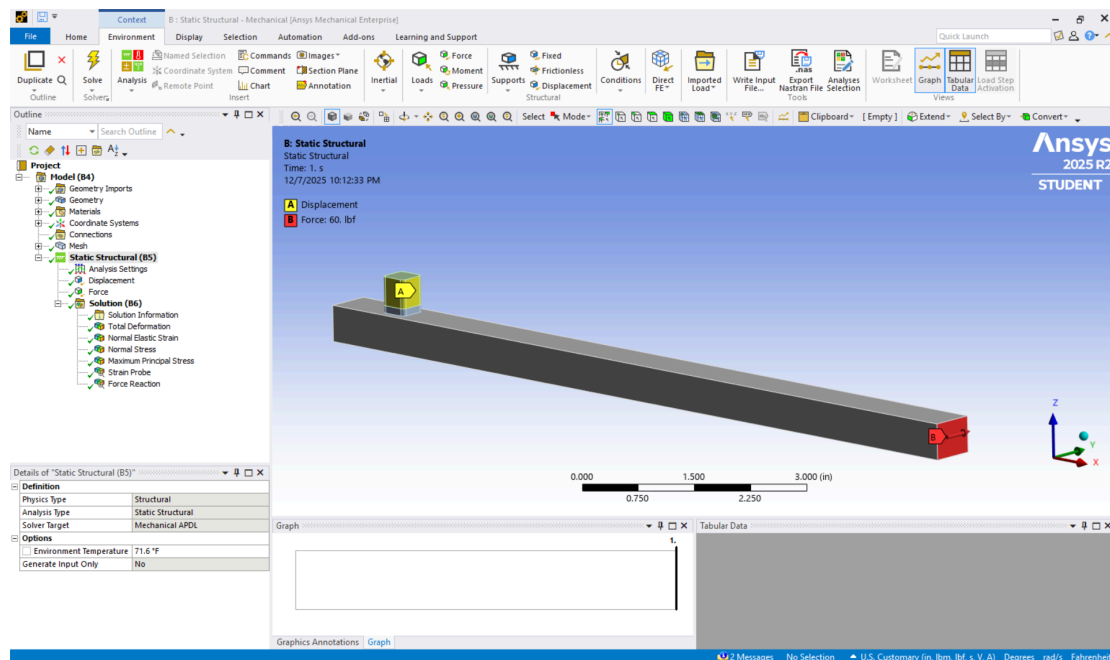
1)



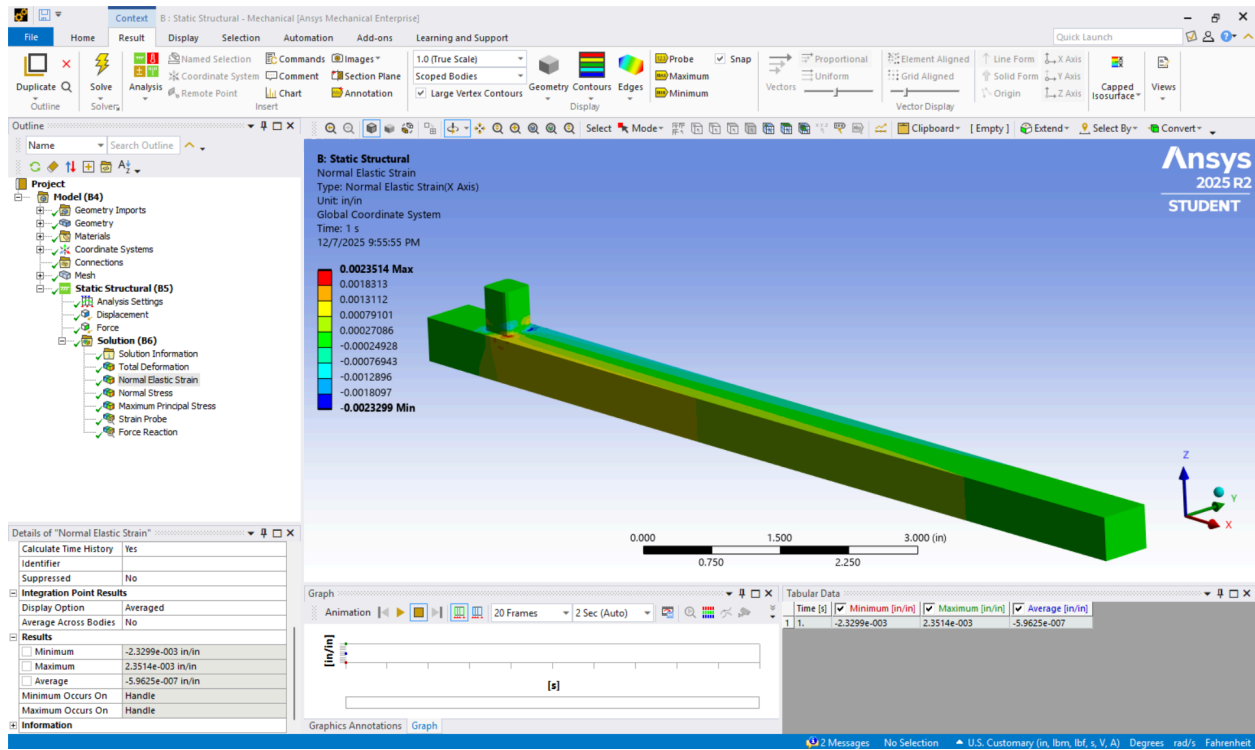
2)

The material I used is called “Titanium, alpha-beta alloy, Ti-6Al-4V, cast” on Granta. This material has a Young’s modulus of $16.7E+6$, a tensile strength of $126E+3$, a fracture toughness of $94.2E+3$, and a fatigue strength of $92E+3$ for 10^7 cycles. This titanium alloy is between the stiffness and tensile strength of steel and aluminum, but it has a very high fracture toughness and fatigue strength. This means that it is not as stiff as steel but stiffer than aluminum, weaker than steel but stronger than aluminum, is more resistant to crack growth than both, and can be used for longer without failing.

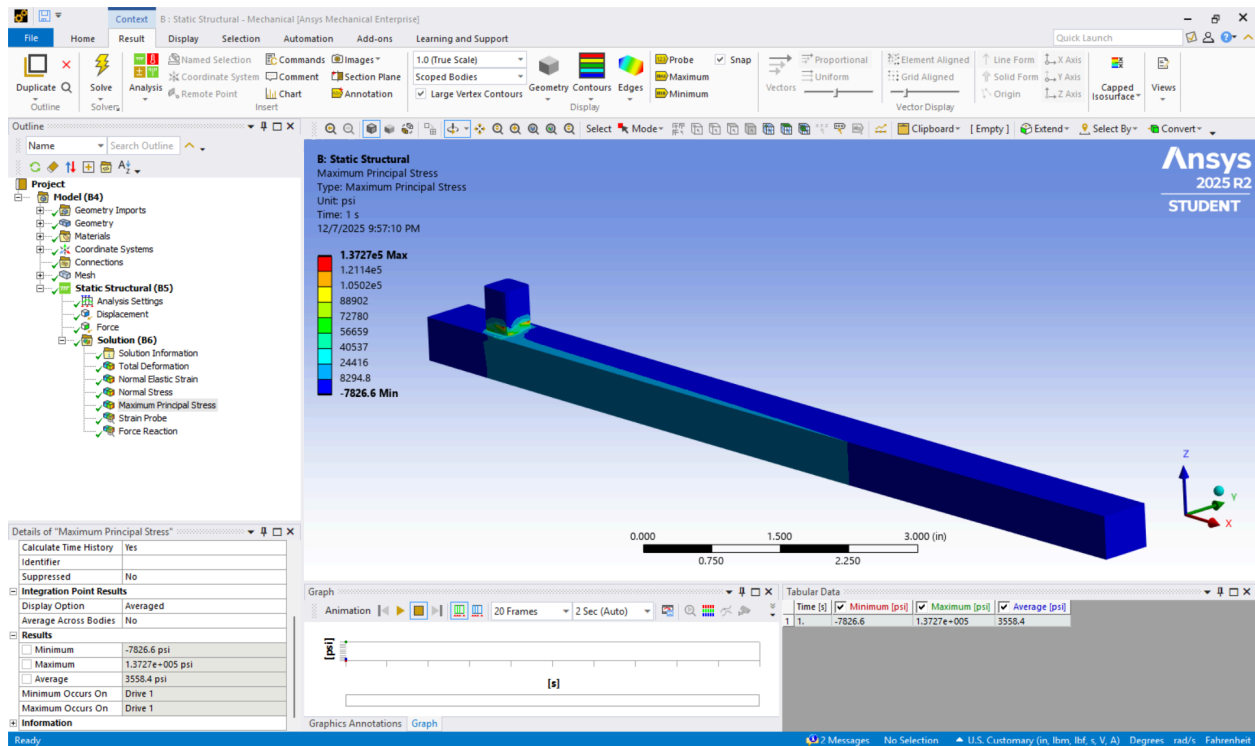
3)



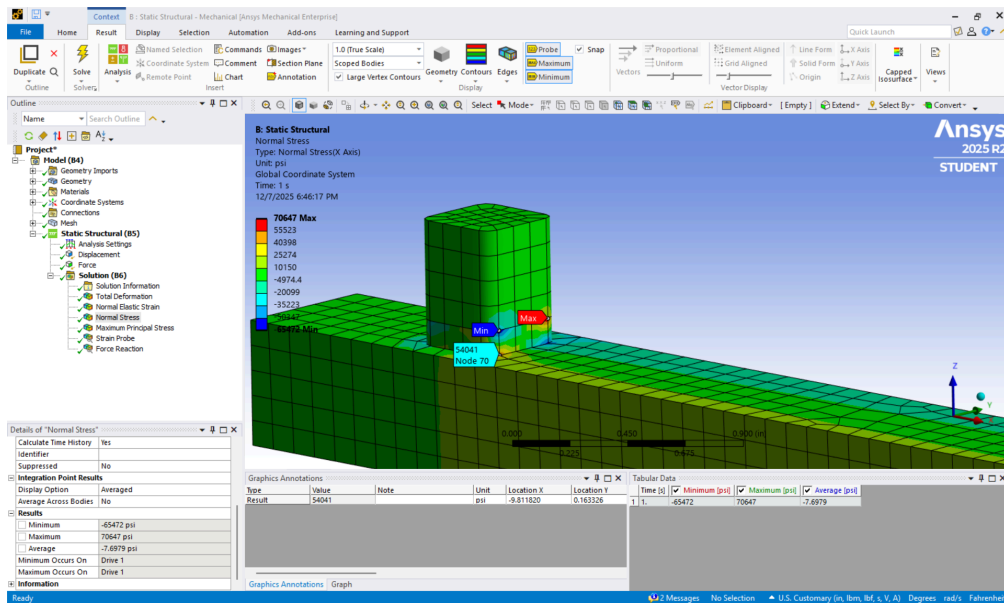
4)



5)

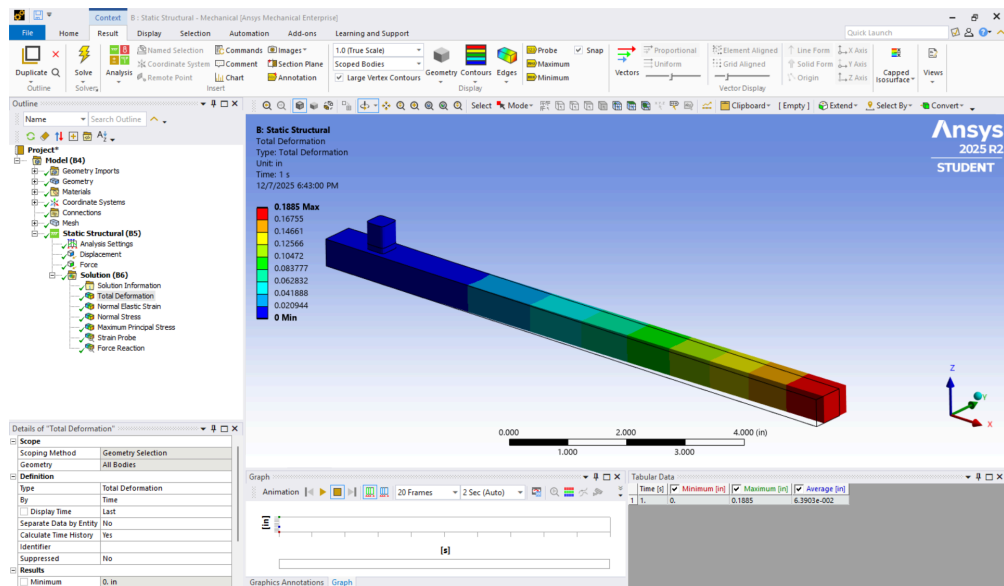


6)

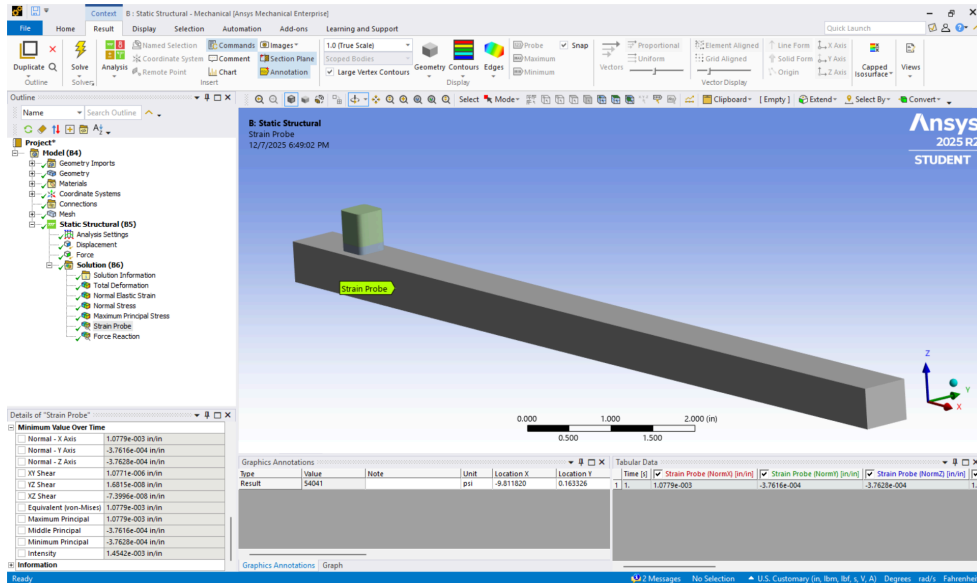


Max normal stress: 70647 psi

Max normal stress on body at probe: 54041 psi



Deflection at load point: 0.1885 in



Strain at probe:
1.0779E-3

7)

From the FEM, the measured strain at the strain gauge is 1.0779E-3. This value multiplied by 1000 results in the torque wrench sensitivity in mV/V. This value is 1.0779 mV/V.

8)

I used a half-bridge strain gauge for this system. I would use the SGD-3S/120-LY13 linear strain gauge which is 0.260"x0.130". This geometry would be able to fit on the side of the wrench, and also works to measure the x-direction strain (FEM coordinate system) of the wrench.